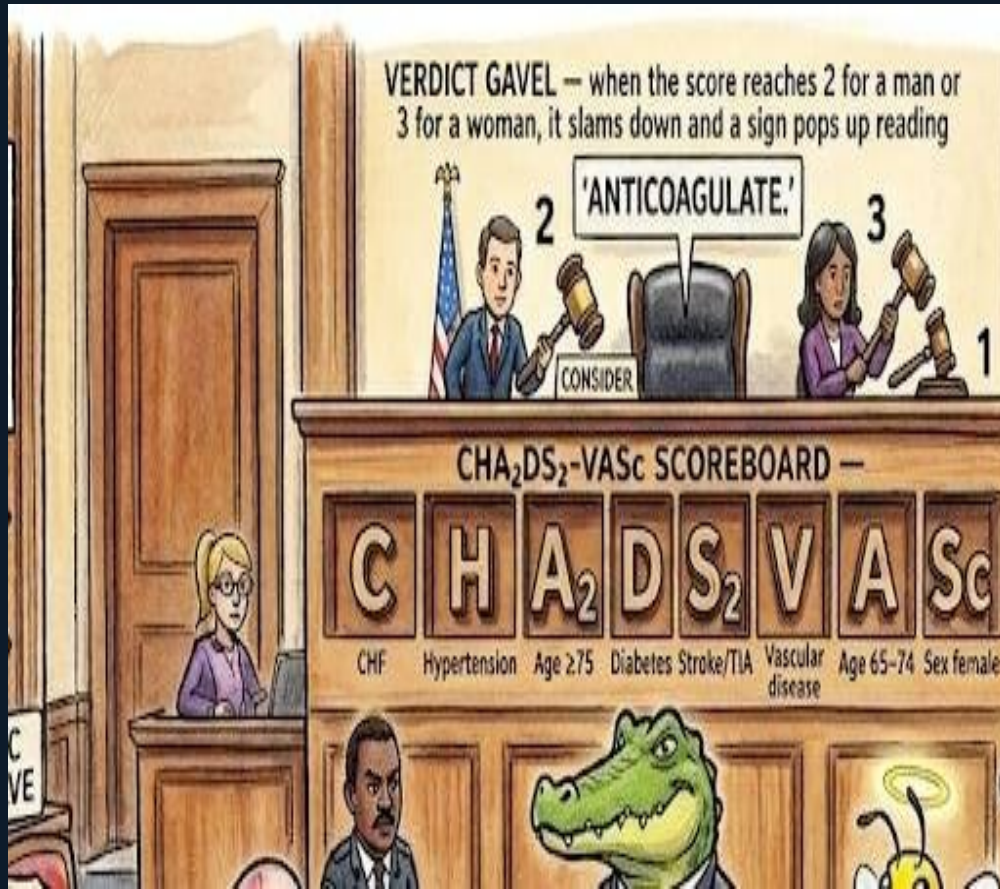


Scene C: AF Anticoagulation



AF Courtroom • MUST MEMORIZE • 4 Board Questions



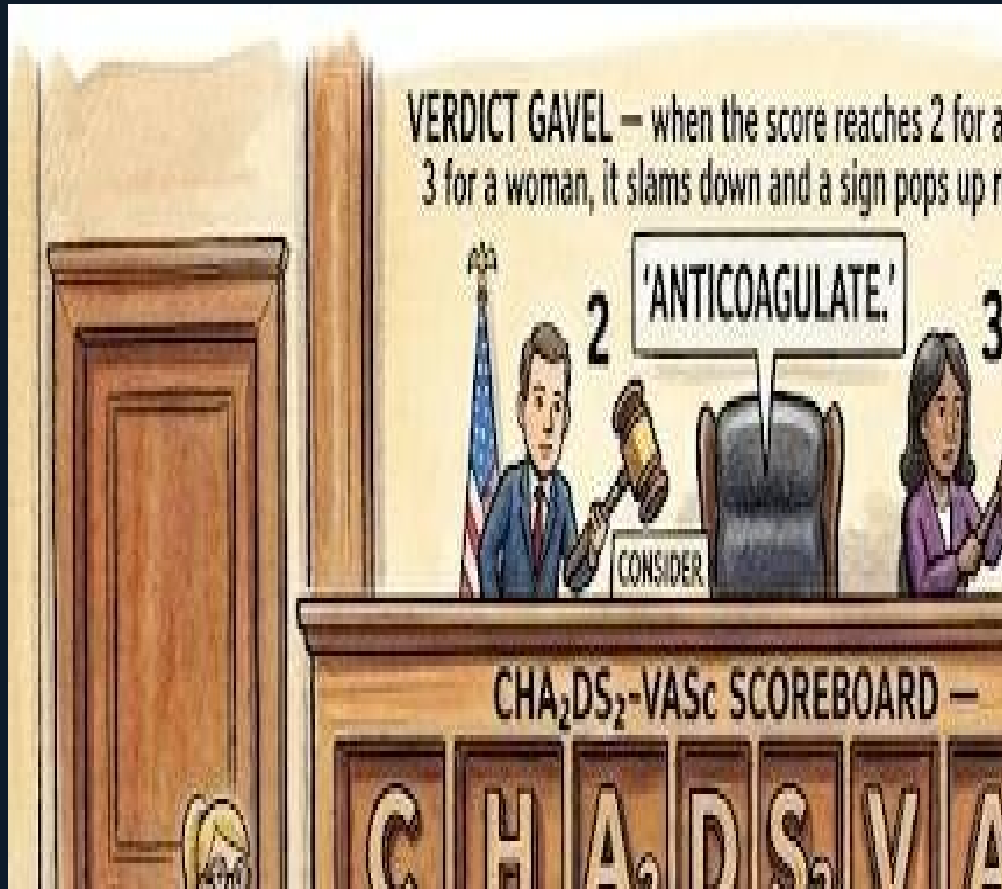
Scene C — AF Anticoagulation | Arrhythmias Board Prep

CHA₂DS₂-VASc Scoreboard

The judge's bench holds 8 carved compartments: C-H-A₂-D-S₂-V-A-Sc. Two gavels — male judge at score 2, female judge at score 3 — slam when the threshold is reached.

CHA₂DS₂-VASc assigns 1 point each for CHF, Hypertension, Diabetes, Vascular disease, Age 65–74, and female Sex. A₂ (Age ≥75) and S₂ (Stroke/TIA) each score 2 points.

Anticoagulate: score ≥2 in men, ≥3 in women. Score 1 in men (not just sex category): consider anticoagulation. Sex category (female) alone at score 1 does not warrant anticoagulation.



Scene C — AF Anticoagulation | Arrhythmias Board Prep

The Gavel Thresholds

Male judge with gavel labeled '2' — slams at $\text{CHA}_2\text{DS}_2\text{-VASc} \geq 2$. Female judge with gavel labeled '3' — slams at ≥ 3 . A 'CONSIDER' label sits below.

The A_2 and S_2 tokens are visibly larger and heavier than the others because they are worth 2 points. A patient aged 75 with prior stroke already scores 4 points — automatic anticoagulation.

Board trap: 'a 65-year-old woman with no other risk factors (score = 1 for sex) needs anticoagulation' — wrong. Female sex alone at score 1 does not trigger anticoagulation; she needs at least one additional risk factor.



The Gator — Dabigatran

A crocodile in a business suit (dabigatran) has a broken kidney chained to its ankle with a red X. Label: 'CrCl <30 → CONTRAINDICATED.'

Dabigatran is the only DOAC that is substantially renally cleared (80% renal excretion). In severe renal impairment (CrCl <30 mL/min), dabigatran accumulates and the bleeding risk becomes unacceptable. It is contraindicated.

Board trap: giving dabigatran to a patient with CrCl 20 — the most tested DOAC renal error. When renal function is impaired, choose apixaban (safest) or adjust rivaroxaban dose.



Scene C — AF Anticoagulation | Arrhythmias Board Prep

The Bee — Apixaban in CKD

A bee character (api = bee → apixaban) wears a safety halo. Label: 'SAFEST IN CKD.' A nearby cracked kidney has no red X — the bee can coexist with renal impairment.

Apixaban has the least renal clearance (~27%) of the DOACs. It can be used down to CrCl 15 mL/min in most guidelines, with dose reduction if 2 of 3 criteria: age ≥ 80 , weight ≤ 60 kg, Cr ≥ 1.5 mg/dL.

Board trap: 'all DOACs are contraindicated in CKD' — wrong. Apixaban is preferred in advanced CKD. Rivaroxaban is used down to CrCl 15 with dose reduction. Only dabigatran is contraindicated at CrCl < 30 .



Scene C — AF Anticoagulation | Arrhythmias Board Prep

Rivaroxaban — Dose Adjustment

A flowing river character (river-oxaban = rivaroxaban) carries a dosing badge: '20 mg → 15 mg if CrCl 15–50.'

Rivaroxaban standard dose in AF is 20 mg once daily with dinner. When CrCl falls to 15–50 mL/min, reduce to 15 mg once daily. Below 15 mL/min: avoid.

Board trap: continuing rivaroxaban 20 mg in a patient with CrCl 35 without dose reduction. The dose adjustment is mandatory and board-tested.



Warfarin — Valvular AF Only

An elderly magistrate (warfarin) sits chained to two objects: a mechanical heart valve (metallic clicking device) and a rheumatic mitral valve (scarred, stiff). Red Xs are stamped on all DOACs from this section.

Warfarin is the only anticoagulant for valvular AF: mechanical heart valves and rheumatic mitral stenosis. In trials, DOACs performed worse than warfarin in mechanical valves (RE-ALIGN trial: increased thromboembolic events with dabigatran).

Board trap: 'use a DOAC in a patient with AF and a mechanical mitral valve' — wrong. Warfarin only. No exceptions.



The Renal Bailiff

A damaged kidney ($\text{CrCl} = 20$ written on it) approaches the gator/dabigatran. A stern bailiff immediately blocks it with a red X barrier. Badge: 'RENAL FAILURE → NO DABIGATRAN.'

This is the most-tested DOAC selection question: patient with AF, renal failure, needs anticoagulation — dabigatran is wrong. The bailiff physically prevents the match.

Correct answer: apixaban (dose-adjusted if meets criteria). This distinction is the #1 DOAC board trap in the entire arrhythmia chapter.

Scene C — Board Trap Master Summary

CHA₂DS₂-VASc: anticoag ≥2 male, ≥3 female

A₂ and S₂ = 2 pts each; others = 1 pt

Dabigatran CI in CrCl <30

80% renally cleared; accumulates and bleeds

Rivaroxaban: 20→15 mg if CrCl 15–50

Mandatory dose reduction; board-tested

DOACs preferred in non-valvular AF

Over warfarin for most patients

Female sex alone (score 1) → no anticoag

Needs ≥1 additional risk factor

Apixaban safest in CKD

~27% renal clearance; use to CrCl 15

Mechanical valve + rheumatic MS → warfarin only

DOACs contraindicated; RE-ALIGN trial

Apixaban dose reduction: 2 of 3 criteria

Age ≥80, wt ≤60 kg, Cr ≥1.5